PATRICK T. ANDERSON

ptander01@gmail.com | Clemson, SC | github.com/Ptander01 | pta-portfolio.vercel.app

PROFESSIONAL SUMMARY

Geospatial Data Scientist and Spatial Engineer with 10+ years building production-grade geospatial pipelines, remote sensing workflows, spatial databases, and analytical dashboards across AI infrastructure, federal healthcare, and environmental science. Background in agricultural and environmental science provides deep domain fluency across hydrology, land-cover analysis, vegetation modeling, and utility infrastructure. Known for owning research and engineering projects end-to-end — from problem framing, data acquisition, and methods design through analysis, QA/QC validation, visualization, and stakeholder delivery — and for communicating complex technical findings clearly to both technical teams and non-technical leadership. Demonstrated ability to acquire new tools rapidly under production pressure: transitioned from GIS analyst to full-stack data engineer with no prior software development background, building a 233,000-line C-suite-deployed system in under two months using AI-assisted development workflows.

TECHNICAL SKILLS

GIS & Spatial Analysis: ArcGIS Pro, ArcGIS Online/Portal, ArcGIS Enterprise, QGIS, Google Earth Engine, spatial modeling, network analysis, terrain analysis, suitability modeling, parcel intelligence

Remote Sensing & Earth Obs: Commercial satellite imagery, multitemporal change detection, land-cover classification, UAV/photogrammetry (Pix4D, Agisoft), LiDAR processing (HTC), multispectral analysis, construction monitoring

Python & Data Engineering: GeoPandas, Rasterio, GDAL, Shapely, Pandas, NumPy, Scipy, ArcPy, Xarray; ETL pipelines, schema harmonization, QA/QC automation, large-scale dataset integration, reproducible workflows

Spatial Databases & Formats: PostGIS, PostgreSQL, enterprise geodatabases, GeoJSON, GeoTIFF, Shapefile, NetCDF, coordinate reference systems, projection management

ML & Statistics: Supervised and unsupervised ML, geographically weighted regression, spatial statistics, feature engineering, model evaluation, descriptive and inferential statistics; published across three peer-reviewed works

AI & Agentic Development: AI-assisted development workflows, LLM integration, agentic pipeline design, prompt engineering, 500K+ lines production Python and React via AI-assisted development

Visualization & Dashboards: ArcGIS Dashboards, interactive web maps, React, Three.js, Recharts, executive briefing design, data storytelling, technical-to-non-technical communication

Cloud & Infrastructure: AWS (S3, EC2, Batch), Docker, Git/GitHub, CI/CD, automated testing, API integration, SQL, JSON/YAML

PROFESSIONAL EXPERIENCE

Meta (AI Research Labs) — 2025 – 2026

Lead Spatial Data Scientist / Remote Sensing Researcher

- Architected a 233,000-line production-grade geospatial data system covering the full analytical lifecycle — ingestion, schema harmonization, spatial and statistical analysis, confidence scoring, multi-resolution layer publishing, and automated diagnostic reporting — unifying six heterogeneous global datasets (35,000+ records) and achieving >90% reduction in manual processing time.
- Invented a Universal Consensus ID (UCID) spatial clustering algorithm to resolve conflicting data center identities across sources; built a companion 41-script QA validation suite generating daily HTML diagnostic reports tracking accuracy and confidence across the full pipeline.
- Developed a global historical satellite imagery viewer enabling multitemporal construction monitoring, footprint delineation, infrastructure classification, and change detection across hundreds of hyperscale data center sites worldwide.
- Built an interactive ArcGIS dashboard integrating global consensus polygons, OSM infrastructure layers, 165M-record parcel datasets, and market-signal heat maps with site-level filters, queries, and construction timelines — used daily by C-suite and strategy leadership.
- Produced recurring executive analytic briefs on AI infrastructure expansion, competitive siting trends, and construction pacing, synthesizing remote sensing, parcel, and market intelligence into leadership-ready insights.
- Transitioned from GIS analyst to full-stack data engineer with no prior software development background, leveraging AI-assisted development tooling to build the full system from first line of code to C-suite deployment in under two months; produced 500,000+ lines of production-grade Python and React across the engagement.

Booz Allen Hamilton — 2021 – 2025

Lead Spatial Data Scientist

- Led geospatial analytics and spatial modeling for national VA healthcare infrastructure, delivering decision-support tools and visualizations used by federal program leadership to guide facility planning, coverage optimization, and policy decisions across hundreds of care centers.
- Built ML-powered spatial forecasting models incorporating geographically weighted regression, demographic feature engineering, historical utilization patterns, and socioeconomic indices; optimized facility coverage using Voronoi-based spatial algorithms and

origin-destination cost-matrix datasets.
- Modernized GIS facility maps and data infrastructure for 100+ federal healthcare centers; developed reusable geospatial workflows, enterprise geodatabases, and ArcGIS dashboards improving data accuracy and accessibility across national programs.
- Presented research findings to federal executive leadership, national clinical staff, and GIS professionals — including three consecutive VHA Executive Chief's Summits and the Annual Corporate Data Warehouse Insights Day.

College of Agriculture, Forestry & Life Sciences — Clemson University 2016 – 2020

GIS and Remote Sensing Researcher

- Performed landscape-scale spatial analysis across 90,000+ km² for SCDNR integrating elevation, land cover, forest density, soils, and NHD hydrology into multi-layer environmental decision-support models.
- Modeled PV site suitability across 88 agricultural sites using UAV and satellite imagery, irradiance modeling, and utility infrastructure constraints — published in Drones (MDPI, 2020) and basis of MS thesis on rural energy supply management.
- Automated distributed LiDAR processing pipelines (HTC) and developed Python tools for fluvial geomorphology analysis and feature sinuosity measurement from USGS 3DHP, NHD, and NHD+ datasets; researched canopy structure using UAV photogrammetry (Pix4D, Agisoft).

Clemson Cooperative Extension 2015 – 2020

Environmental Data Analyst

- Led a $3.4M federal energy grant project completing 150+ agricultural engineering assessments; modeled thermodynamics, fluid mechanics, and climatology for agricultural engineering applications.
- Developed ROI, time-series utility, and cost-benefit financial models; authored 300+ technical reports providing actionable engineering and energy insights to project managers, engineers, and agribusiness clients.

EDUCATION

Master of Science in Environmental Science GPA: 3.7 / 4.0

Clemson University — Specialization: GIS, Remote Sensing, Photogrammetry & Statistics 2020

Thesis: Evaluating Photovoltaics in a Peak-Shaving Supply Management Role to Rural Communities

Publication: Koc, A. & Anderson, P. “Estimating Rooftop Areas of Poultry Houses For Solar Energy Applications Using UAV and Satellite Images.” Drones, MDPI (2020) 4, 76.

Bachelor of Science in Agricultural Science GPA: 3.5 / 4.0

Clemson University 2014

Publication: Chastain, J. & Anderson, P. “Analysis of Available Efficiency and Performance Data for Axial Flow Agricultural Ventilation.” ASABE Annual International Meeting (2017) 1700116.